# Answers and talk notes

Divide Multi-Digit Numbers Using an Algorithm

### What your student is learning

Tonight we take the long-division steps from Part 1 and use them on a real staffing problem — how many teachers a district has to hire.

## Answers

Ask your student to explain before you look.

1. **B.** 20. 40 ÷ 2 = 20 with one zero left over, so 400 ÷ 20 = 20.
2. **A.** The number of teachers. The total is split into groups of 20, so the quotient 420 is the number of teachers. So the answer is: The number of teachers.
3. **C.** 120. 15 × 100 = 1,500 and 15 × 20 = 300, so 1,800 ÷ 15 = 100 + 20 = 120.
4. **D.** 30. 600 ÷ 20 = 30 adults.
5. **A.** 10. Now: 600 ÷ 30 = 20 adults. Goal: 600 ÷ 20 = 30 adults. 30 − 20 = 10 more.

### Talk about it

What does each quotient count, and why do you subtract at the end?

### Home mission

Say the plan out loud: one teacher for every 20 students, then one for every 15. Which plan needs more teachers? How do you know before you divide?

### When they are stuck, ask

- What does this quotient count — students, or adults?
- How many groups of that size fit into the total?
- Does the question ask for the new total, or for how many MORE?

### Try not to say

- If it will not divide, just skip that digit.
- Smaller groups means you need fewer adults.
- Just do one division and that is the answer.